ANSWERS

1 (a) *either* energy (stored)/work done represented by area under graph
or energy = <u>average</u> force × extension B1
energy = ½ × 180 × 4.0 × 10^{-2} C1
= 3.6 J .... A1 [3]

(b) (i) *either* momentum before release is zero M1
so sum of <u>momenta</u> (of trolleys) after release is zero A1
or force = rate of change of momentum (M1)
force on trolleys equal and opposite (A1)
or impulse = change in momentum (M1)
impulse on each equal and opposite (A1) [2]

(ii) 1 $M_1V_1 = M_2V_2$ .... B1 [1]

2 $\underline{E} = ½\, M_1V_1^2 + ½\, M_2V_2^2$ .... B1 [1]

(iii) 1 $E_K = ½mv^2$ and $p = mv$ combined to give M1
$E_K = p^2 / 2m$ .... A0 [1]

2 *m* smaller, E_K is larger because *p* is the same/constant M1
so trolley B .... A0 [1]

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2 (a) work done is the force × the distance moved / displacement in the direction of the force
or
work is done when a force moves in the direction of the force B1 [1]

(b) component of weight = 850 × 9.81 × sin 7.5° C1
= 1090 N A1 [2]
(*use of incorrect trigonometric function, 0/2*)

(c) (i) ΣF = 4600 – 1090 = (3510) M1
deceleration = 3510 / 850 A1
= 4.1 ms^{-2} A0 [2]

(ii) $v^2 = u^2 + 2as$
0 = 25^2 + 2 × – 4.1 × *s* C1
s = 625 / 8.2
= 76 m A1 [2]
(*allow full credit for calculation of time (6.05 s) & then s*)

(iii) 1. kinetic energy = ½ mv^2 C1
= 0.5 × 850 × 25^2
= 2.7 × 10^5 J A1 [2]

2. work done = 4600 × 75.7
= 3.5 × 10^5 J A1 [1]

(iv) difference is the loss in potential energy (*owtte*) B1 [1]

3	(a)	(i)	work done equals force × distance moved / displacement in the direction of the force	B1	[1]
		(ii)	power is the rate of doing work / work done per unit time	B1	[1]
	(b)	(i)	kinetic energy $= \frac{1}{2} mv^2$	C1	
			$= 0.5 \times 600\ (9.5)^2$	C1	
			$= 27075$ (J) $= 27$ kJ	A1	[3]
		(ii)	potential energy $= mgh$		
			$= 600 \times 9.81 \times 4.1$	M1	
			$= 24132$ (J)	A1	
			$= 24$ kJ	A0	[2]
		(iii)	work done $= 27 - 24 = 3.0$ kJ	A1	[1]
		(iv)	resistive force $= 3000 / 8.2$ (distance along slope $= 4.1 / \sin 30°$)	C1	
			$= 366$ N	A1	[2]

4	(a)		electrical potential energy (stored) when charge moved and gravitational potential energy (stored) when mass moved	B1	
			due to work done in electric field and work done in gravitational field	B1	[2]
	(b)		work done = force × distance moved (in direction of force) and force $= mg$	M1	
			$mg \times h$ or $mg \times \Delta h$	A1	[2]
	(c)	(i)	$0.1 \times mgh = \frac{1}{2} mv^2$	B1	
			$0.1 \times m \times 9.81 \times 120 = 0.5 \times m \times v^2$	B1	
			$v = 15.3\ \text{m s}^{-1}$	A0	[2]
		(ii)	$P = 0.5\ m\ v^2 / t$	C1	
			$m / t = 110 \times 10^3 / [0.25 \times 0.5 \times (15.3)^2]$	C1	
			$= 3740\ \text{kg s}^{-1}$	A1	[3]

5	(a)	(i)	force is rate of change of momentum	B1	[1]
		(ii)	work done is the product of the force and the distance <u>moved</u> in the direction of the force	B1	[1]
	(b)	(i)	$W = Fs$ **or** $W = mas$ **or** $W = m(v^2 - u^2) / 2$ **or** W = force × distance s	A1	[1]
		(ii)	$as = (v^2 - u^2) / 2$ any subject	M1	
			$W = mas$ hence $W = m(v^2 - u^2) / 2$	M1	
			RHS represents terms of energy **or** with $u = 0$ KE $= \frac{1}{2}mv^2$	A1	[3]
	(c)	(i)	work done $= \frac{1}{2} \times 1500 \times [(30)^2 - (15)^2]$ (=506250)	C1	
			distance = WD / F = 506250 / 3800 = 133 m	A1	[2]
			or $F = ma$ $a = 2.533\ (\text{m s}^{-2})$	C1	
			$v^2 = u^2 + 2as$ $s = 133$ m	A1	
		(ii)	the change in kinetic energy is greater or the work done by the force has to be greater, hence distance is greater (for same force)	A1	[1]
			allow: same acceleration, same time, so greater average speed and greater distance		

6	**(a)**		$V = h \times A$	
			$m = V \times \rho$	B1
			$W = h \times A \times \rho \times g$	B1
			$P = F / A$	B1
			$P = h\rho g$	
			P is proportional to h if ρ is constant (and g)	B1 [4]
	(b)		density changes with height	B1
			hence density is not constant with link to formula	B1 [2]
7	**(a)**		power is the rate of doing work or power = work done / time (taken) or power = energy transferred / time (taken)	B1 [1]
	(b)	**(i)**	as the speed increases drag / air resistance increases	B1
			resultant force reduces hence acceleration is less	B1
			constant speed when resultant force is zero	B1 [3]
			(allow one mark for speed increases and acceleration decreases)	
		(ii)	force from cyclist = drag force / resistive force	B1
			$P = 12 \times 48$	M1
			$P = 576\,\text{W}$	A0 [2]
		(iii)	tangent drawn at speed = $8.0\,\text{m}\,\text{s}^{-1}$	M1
			gradient values that show acceleration between 0.44 to $0.48\,\text{m}\,\text{s}^{-2}$	A1 [2]
		(iv)	$F - R = ma$	C1
			$600 / 8 - R = 80 \times 0.5$ [using $P = 576$] $576 / 8 - R = 80 \times 0.5$	C1
			$R = 75 - 40 = 35\,\text{N}$ $R = 72 - 40 = 32\,\text{N}$	A1 [3]
		(v)	at $12\,\text{m}\,\text{s}^{-1}$ drag is 48 N, at $8\,\text{m}\,\text{s}^{-1}$ drag is 35 or 32 N	
			R / v calculated as 4 and 4 or 4.4	
			and consistent response for whether R is proportional to v or not	B1 [1]
8	**(a)**		loss in potential energy due to decrease in height (as P.E. = mgh)	(B1)
			gain in kinetic energy due to increase in speed (as K.E. = $\frac{1}{2} mv^2$)	(B1)
			special case 'as PE decreases KE increases' (1/2)	
			increase in thermal energy due to work done against air resistance	(B1)
			loss in P.E. equals gain in K.E. and thermal energy	(B1)
				max. 3 [3]
	(b)	**(i)**	kinetic energy $= \frac{1}{2} mv^2$	C1
			$= \frac{1}{2} \times 0.150 \times (25)^2$	C1
			$= 46.875 = 47\,\text{J}$	A1 [3]
		(ii) 1.	potential energy (= mgh) = $0.150 \times 9.81 \times 21$	C1
			loss = KE – mgh = 46.875 – (30.9)	C1
			= 15.97 = 16 J	A1 [3]
		2.	work done = 16 J	
			work done = force $\times$ distance	C1
			$F = 16 / 21 = 0.76\,\text{N}$	A1 [2]

9 **(a)** power = energy / time	C1	
= (force × distance / time) = kg $m^2 s^{-2}$ / s	C1	
= kg $m^2 s^{-3}$	A1	[3]
(b) **(i)** units of L^2: m^2 and units of ρ: kg m^{-3} and units of v^3: $m^3 s^{-3}$	C1	
($C = P / L^2 \rho v^3$) hence units of C: kg $m^2 s^{-3} m^{-2} kg^{-1} m^3 m^{-3} s^3$		
or any correct statement of component units	M1	
argument /discussion / cancelling leading to C having no units	A1	[3]
(ii) power available from wind = $3.5 \times 10^5 \times 100 / 55$ (= 6.36×10^5)	C1	
$v^3 = 3.5 \times 10^5 \times 100 / (55 \times 0.931 \times (25)^2 \times 1.3)$	C1	
$v = 9.4$ m s^{-1}	A1	[3]
(iii) not all kinetic energy of wind converted to kinetic energy of blades	B1	
generator / conversion to electrical energy not 100% efficient / heat produced in generator / bearings etc	B1	[2]
(there must be cause of loss and where located)		

10 (a) (work =) force × distance <u>moved</u> / displacement in the direction of the force OR when a force moves in the direction of the force work is done	B1	[1]
(b) kinetic energy = ½ mv^2	C1	
= <u>½ 0.4 $(2.5)^2$ = 1.25 / 1.3 J</u>	A1	[2]
(c) **(i)** area under graph is work done / work done = ½ Fx	C1	
1.25 = (14 x) / 2	C1	
x = 0.18 (0.179) m [allow x = 0.19 m using kinetic energy = 1.3 J]	A1	[3]
(ii) smooth curve from v = 2.5 at x = 0 to v = 0 at Q	M1	
curve with increasing gradient	A1	[2]

11 **(a)** gravitational PE is energy of a <u>mass</u> due to its position in a <u>gravitational field</u>	B1	
elastic PE energy <u>stored</u> (in an object) <u>due to</u> (a force) changing its shape / deformation / being compressed / stretched / strained	B1	[2]
(b) **(i)** **1.** kinetic energy = ½ mv^2	C1	
= ½ × 0.065 × 16^2 = 8.3(2) J	A1	[2]
2. $v^2 = 2gh$ OR PE = mgh	C1	
$h = 16^2 / (2 \times 9.81)$ = 13(.05) m	A1	[2]
(ii) speed at t = ½ total time = 8 (m s^{-1}) or total t =1.63 or $t_{1/2}$ = 0.815 s	C1	
KE is ¼ or h at $t_{1/2}$ = 9.78 (m)	C1	
and PE is ¾ of max ratio = 3 or ratio = 9.78 / 3.26 = 3	A1	[3]
(iii) time is less because (average) acceleration is greater OR average force is greater	B1	[1]

12 (a) (i)	work (done)/time (taken)	B1	[1]
(ii)	work = force × displacement (in direction of force)	B1	
	power = force × displacement/time (taken) = force × velocity	B1	[2]
(b) (i)	weight = mg	C1	
	$P = Fv = 2500 \times 9.81 \times \sin 9° \times 8.5$ (or use cos 81°)	C1	
	= 33 (32.6) kW	A1	[3]
(ii)	no gain or loss of KE	B1	
	no work (done) against air resistance	B1	[2]
13 (a)	kinetic energy = ½ mv_2	C1	
	$= ½ \times 0.040 \times (2.8)^2 = 0.157$ J or 0.16 J	A1	[2]
(b) (i)	$k = F/x$ or $F = kx$	C1	
	$X_B = 14/800$		
	= 0.0175 m	A1	[2]
(ii)	area under graph = elastic potential energy stored	C1	
	or ½ kx^2 *or* ½ Fx		
	(energy stored =) 0.1225 J less than KE (of 0.16 J)	A1	[2]
14 (a)	GPE: energy of a <u>mass</u> due to its position in a <u>gravitational field</u>	B1	
	KE: energy (a mass has) due to its motion/speed/velocity	B1	[2]
(b) (i) 1.	$KE = \frac{1}{2} mv^2$	C1	
	$= \frac{1}{2} \times 0.4 \times (30)^2$	C1	
	= 180 J	A1	[3]
2.	$s = 0 + \frac{1}{2} \times 9.81 \times (2.16)^2$ *or* $s = (30 \sin 45°)^2/(2 \times 9.81)$	C1	
	= 22.88 (22.9) m = 22.94 (22.9) m	A1	[2]
3.	GPE = mgh	C1	
	$= 0.4 \times 9.81 \times 22.88 = 89.8$ (90) J	A1	[2]
(ii) 1.	KE = initial KE – GPE = 180 – 90 = 90 J	A1	[1]
2.	(horizontal) velocity is not zero/(object) is still moving/answer explained in terms of conservation of energy	B1	[1]

15 (a)	work done is the product of force and the distance <u>moved</u> in the direction of the force *or* product of force and displacement in the direction of the force	B1	[1]
(b) (i)	work done equals the <u>decrease</u> in GPE – <u>gain</u> in KE	B1	[1]
(ii) 1.	distance = area under line	C1	
	= (7.4 × 2.5) / 2 = 9.3 m (9.25 m)	M1	[2]
or	acceleration from graph a = 7.4 / 2.5 (= 2.96)	(C1)	
	and equation of motion $(7.4)^2 = 2 \times 2.96 \times s$ gives s = 9.3 (9.25) m	(A1)	
2.	kinetic energy = $\frac{1}{2}mv^2$	C1	
	$= \frac{1}{2} \times 75 \times (7.4)^2$	C1	
	= 2100 J	A1	[3]
3.	potential energy = mgh	C1	
	h = 9.3 sin 30°	C1	
	PE = 75 × 9.81 × 9.3 sin 30° = 3400 J	A1	[3]
4.	work done = energy loss	C1	
	R = (3421 – 2054) / 9.3	C1	
	= 150 (147) N	A1	[3]

16 (a) (i)	change in kinetic energy = ½mv^2	C1	
	= 0.5 × 25 × $(0.64)^2$ = 5.1(2) J	A1	[2]
(ii)	zero	A1	[1]
(iii)	(–) 5.1(2) J	A1	[1]
(b) (i)	PE = mgh	C1	
	= 350 × 0.64 × 25	C1	
	= 5600 J	A1	[3]
	(If full length used allow 1/3)		
(ii)	P = Fv or gain in PE / t, E_P / t or work done / t, W / t	C1	
	= 350 × 0.64 or 5600 / 25		
	= 220 (224) W	A1	[2]

17 (a)	(power =) work done / time (taken) or rate of work done	A1	[1]
(b) (i)	$F - R = ma$	C1	
	F = 1500 × 0.82 + 1200	C1	
	= 2400 (2430) N	A1	[3]
(ii)	$P = Fv$	C1	
	= (2430 × 22) = 53 000 (53 500) W	A1	[2]
(c)	(there is maximum power from car and) resistive force = force produced by car hence no acceleration		
or	suggestion in terms of power produced by car and power wasted to overcome resistive force	B1	[1]

18 (a) (i)	(work =) force × distance <u>moved</u> <u>in the direction of the force</u>.	B1	
(ii)	the energy <u>stored</u> (in an object) due to extension / compression / change of shape	B1	
(b) (i)	$E_K = ½mv^2$	C1	
	$= 0.5 \times 0.40 \times 0.30^2$		
	$= 1.8 \times 10^{-2}$ (J)	A1	
(ii)	(change in) kinetic energy = work done on spring / (change in) elastic potential energy	C1	
	$1.8 \times 10^{-2} = ½ \times F \times 0.080$	C1	
	$F_{MAX} = 0.45$ (N)	A1	
(iii)	$a = F / m = 0.45 / 0.40$		
	$= 1.1$ (m s^{-2})	A1	
(iv)	**1.** constant velocity / resultant force is zero, so in equilibrium	B1	
	2. decelerating / resultant force is not zero, so not in equilibrium	B1	
(c)	curved line from the origin	M1	
	with decreasing gradient	A1	
19 (a)	the energy (stored) in a body due to its extension/compression/deformation/ change in shape/size	B1	[1]
(b) (i)	two values of F/x are calculated which are the same e.g. 10.4 / 40 = 0.26 and 6.5 / 25 = 0.26	B1	
	or		
	ratio of two forces and the ratio of the corresponding two extensions are calculated which are the same e.g. 5.2 / 10.4 = 0.5 and 20 / 40 = 0.5	(B1)	
	or		
	gradient of graph line calculated and coordinates of one point on the line used with straight line equation $y = mx + c$ to show $c = 0$	(B1)	
	(so) force is proportional to extension (and so Hooke's law obeyed)	B1	[2]
(b) (ii)	**1.** $k = F / x$ or k = gradient	C1	
	gradient or values from a single point used e.g. $k = 10.4 / (40 \times 10^{-2})$		
	$k = 26$ N m^{-1}	A1	[2]
	2. work done = area under graph or $½Fx$ or $½(F_2 + F_1)(x_2 - x_1)$ or $½kx^2$ or $½k(x_2^{\,2} - x_1^{\,2})$	C1	
	$= ½ \times 10.4 \times 0.4 - ½ \times 5.2 \times 0.2$ or $½ \times (5.2 + 10.4) \times 20 \times 10^{-2}$ or $½ \times 26 \times (0.4^2 - 0.2^2)$	C1	
	= 1.6 J	A1	[3]
(c)	remove the force and the spring goes back to its original length	B1	[1]

20 (a)	acceleration = change in velocity / time (taken) or rate of change of velocity	B1	[1]
(b) (i)	$v = 0 + at$ or $v = at$	C1	
	(a = 36/19 =) 1.9 (1.8947) m s^{-2}	A1	[2]
(ii)	$s = ½(u + v)t$ or $s = v^2/2a$ or $s = ½at^2$		
	= ½ × 36 × 19 = 36^2/(2 × 1.89) = ½ × 1.89 × 19^2		
	= 340 m (342 m/343 m/341 m)	M1	[1]
(iii)	**1.** (ΔKE =) ½ × 95 × (36)^2	C1	
	= 62 000 (61 560) J	A1	[2]
	2. (ΔPE =) 95 × 9.81 × 340 sin 40° or 95 × 9.81 × 218.5	C1	
	= 200 000 J	A1	[2]
(iv)	work done (by frictional force) = ΔPE – ΔKE		
	or		
	work done = 200 000 – 62 000 (values from **1b(iii) 1.** and **2.**)	C1	
	(frictional force = 138 000/340 =) 410 (406) N [420 N if full figures used]	A1	[2]
(v)	$-ma = mg \sin 20° - f$ or $ma = -mg \sin 20° + f$	C1	
	–95 × 3.0 = 95 × 3.36 – f		
	f = 600 (604) N	A1	[2]

21 (a)	(gravitational potential energy is) the energy/ability to do work of a <u>mass</u> that it has or is stored due to its position/height in a gravitational field	B1	
	kinetic energy is energy/ability to do work a object/body/mass has due to its speed/velocity/motion/movement	B1	[2]
(b) (i)	$s = [(u + v)t]/2$ or acceleration = 9.8/9.75 (using gradient)	C1	
	= [(7.8 + 3.9) × 0.4]/2 or $s = 3.9 \times 0.4 + \frac{1}{2} \times 9.75 \times (0.4)^2$	C1	
	s = 2.3(4) m	A1	[3]
(ii)	$a = (v - u)/t$ or gradient of line	C1	
	= (7.8 – 3.9)/0.4 = 9.8 (9.75) m s^{-2} (allow ± $\frac{1}{2}$ small square in readings)	A1	[2]
(iii)	KE = $\frac{1}{2}mv^2$	C1	
	change in kinetic energy = $\frac{1}{2}mv^2 - \frac{1}{2}mu^2$		
	= $\frac{1}{2} \times 1.5 \times (7.8^2 - 3.9^2)$	C1	
	= 34 (34.22) J	A1	[3]
(c)	work done = force × distance (moved) or Fd or Fx or mgh or mgd or mgx	M1	
	= 1.5 × 9.8 × 2.3 = 34 (33.8) J (equals the change in KE)	A1	[2]

22 (a) $\Delta E = mg\Delta h$ — C1

$= 0.030 \times 9.81 \times (-)0.31$

$= (-)0.091$ J — A1 [2]

(b) $E = ½mv^2$ — C1

(initial) $E = ½ \times 0.030 \times 1.3^2$ (= 0.0254) — C1

$0.5 \times 0.030 \times v^2 = (0.5 \times 0.030 \times 1.3^2) + (0.030 \times 9.81 \times 0.31)$ so $v = 2.8\,\text{m s}^{-1}$

or

$0.5 \times 0.030 \times v^2 = (0.0254) + (0.091)$ so $v = 2.8\,\text{m s}^{-1}$ — A1 [3]

(c) (i) $0.096 = 0.030(v + 2.8)$ — C1

$v = 0.40\,\text{m s}^{-1}$ — A1 [2]

(ii) $F = \Delta p / (\Delta)t$ *or* $F = ma$

$= 0.096 / 20 \times 10^{-3}$ *or* $0.030\,(0.40 + 2.8) / 20 \times 10^{-3}$ — C1

$= 4.8$ N — A1 [2]

(d) kinetic energy (of ball and wall) decreases/changes/not conserved, so inelastic

or

(relative) speed of approach (of ball and wall) not equal to/greater than (relative) speed of separation, so inelastic. — B1 [1]

(e) force = work done / distance moved

$= (0.091 - 0.076) / 0.60$ — C1

$= 0.025$ N — A1 [2]

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23 (a) $v = u + at$ — A1

$v = 9.6 - (9.81 \times 0.37) = 6.0\,\text{m s}^{-1}$

(b) $s = ½ \times (9.6 + 6.0) \times 0.37$ — C1

or $6.0^2 = 9.6^2 - (2 \times 9.81 \times s)$

or $s = (9.6 \times 0.37) - (½ \times 9.81 \times 0.37^2)$

or $s = (6.0 \times 0.37) + (½ \times 9.81 \times 0.37^2)$

$s = 2.9$ m — A1

(c) (i) $(\Delta)E = mg(\Delta)h$ — C1

$\Delta E = 0.056 \times 9.81 \times 2.9$ — A1

$= 1.6$ J

(ii) $E = ½mv^2$ — C1

$\Delta E = ½ \times 0.056 \times (6.0^2 - 3.8^2)$ — A1

$= 0.60$ J

(d) force on ball (by ceiling) equal to force on ceiling (by ball) — M1

and opposite (in direction) — A1

(e) (p =) mv **or** 0.056×6.0 **or** 0.056×3.8 — C1

change in momentum $= 0.056 \times (6.0 + 3.8)$ — A1

$= 0.55$ N s

(f) resultant force = 0.55/0.085 (= 6.47 N) — C1

force by ceiling $= 6.47 - (0.056 \times 9.81)$ — A1

$= 5.9$ N

24 **(a)** **(i)** distance in a specified direction (from a point) **B1**

(ii) change in velocity / time (taken) **B1**

(b) **(i)** vertical component of velocity = $(5.5^2 - 4.6^2)^{1/2} = 3.0$ (m s^{-1}) **A1**

or

$5.5 \cos \theta = 4.6$ (so $\theta = 33.2°$) **and** $5.5 \sin 33.2° = 3.0$ (m s^{-1})

(ii) $s = ut + ½at^2$ **C1**

$0 = (3.0 \times t) - (½ \times 9.81 \times t^2)$

or

$v = u + at$

$-3.0 = 3.0 - 9.81t$

$t = 0.61$ s **A1**

(iii) $d = 4.6 \times 0.61$ **A1**

$= 2.8$ m

(iv) $E = ½mv^2$ **C1**

ratio = $(½ \times m \times 4.6^2) / (½ \times m \times 5.5^2)$ **C1**

or

ratio = $(½ \times m \times 5.5^2 - m \times 9.81 \times 0.459) / (½ \times m \times 5.5^2)$

ratio = 0.70 **A1**

(c) straight line from positive value of v_y at $t = 0$ to negative value of v_y **M1**

straight line ends at $t = T$ and final magnitude of v_y greater than initial magnitude of v_y **A1**

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25 **(a)** **(i)** work (done) / time (taken) **B1**

(ii) energy of a mass due to its position in a gravitational field **B1**

(b) **(i)** $P = Fv$ **C1**

$= 2.0 \times 10^3 \times 45$ **A1**

$= 9.0 \times 10^4$ W

(ii) **1.** $W = (2.0 \times 10^3) \times (45 \times 3.0 \times 60)$ **or** $W = 9.0 \times 10^4 \times 3.0 \times 60$ **C1**

$W = 1.6 \times 10^7$ J **A1**

2. $(\Delta)E_P = mg(\Delta)h$ **C1**

$= 1200 \times 9.81 \times 3.3 \times 3.0 \times 60$ **A1**

$= 7.0 \times 10^6$ J

3. $W = 1.6 \times 10^7 - 7.0 \times 10^6$ **A1**

$= 9.0 \times 10^6$ J

(iii) force $= (9.0 \times 10^6) / (45 \times 3.0 \times 60)$ **A1**

$= 1.1 \times 10^3$ N

(iv) constant velocity so no resultant force **B1**

no resultant force so in equilibrium **B1**

26	(a)	$v_x = (6.0^2 - 4.8^2)^{1/2} = 3.6$ (m s^{-1})	A1
		or	
		$6.0 \sin\theta = 4.8$ (so $\theta = 53.1°$) **and** $v_x = 6.0 \cos 53.1° = 3.6$ (m s^{-1})	
	(b) (i)	straight line from (0, 4.8) to (0.49, 0)	M1
		straight line continues with same slope to (0.98, –4.8) (labelled Y)	A1
	(ii)	a horizontal line	M1
		from (0, 3.6) to (0.98, 3.6) (labelled X)	A1
	(c)	$s = ut + ½at^2$	C1
		$= (4.8 \times 0.49) + (½ \times -9.81 \times 0.49^2)$	
		or	
		$s = ½(u + v)t$ **or** area under graph	
		$= ½ \times (4.8 + 0) \times 0.49$	
		or	
		$s = vt - ½at^2$	
		$= ½ \times 9.81 \times 0.49^2$	
		or	
		$v^2 = u^2 + 2as$	
		$s = 4.8^2 / (2 \times 9.81)$	
		$s = 1.2$ m	A1
	(d)	$(\Delta)E = mg(\Delta)h$	C1
		$E = ½mv^2$	C1
		ratio $= (½ \times m \times 3.6^2) / (m \times 9.81 \times 1.2)$	C1
		or	
		ratio $= [(½ \times m \times 6.0^2) - (m \times 9.81 \times 1.2)] / (m \times 9.81 \times 1.2)$	
		or	
		ratio $= (½ \times m \times 3.6^2) / (½ \times m \times 4.8^2)$	
		ratio = 0.56	A1
	(e)	(force due to) air resistance acts in opposite direction to the velocity	B1
		or	
		(with air resistance, average) resultant force is larger (than weight)	

27 (a) energy (of a mass/body/object) due to motion/speed/velocity **B1**

(b)(i) $E = ½mv^2$ **C1**

$480 = ½ \times m \times 80^2$ so $m = 0.15$ kg **A1**

(ii) 1. $E = mgh$ **or** $\Delta E = mg\Delta h$ **C1**

$= 0.15 \times 9.81 \times 210$ **A1**

$= 310$ J

2. work done $= 480 - 310$ **A1**

$= 170$ J

(iii) work done = Fs **C1**

force $= 170/210$ **A1**

$= 0.81$ N

(iv) curved line from positive value on v-axis to (T, 0) **M1**

magnitude of gradient decreases **A1**

(v) as shell rises force decreases and as shell falls force increases **B1**

as shell rises force is downward and as shell falls force is upward **B1**

or

as shell rises the force decreases and is downward **(B1)**

as shell falls the force increases and is upward **(B1)**

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28 (a)(i) $E = ½Fx$ **or** $E = ½kx^2$ **or** E = area under graph **C1**

$E = ½ \times 4.0 \times 0.32 = 0.64$ J **or** $E = ½ \times 12.5 \times (0.32)^2 = 0.64$ J **A1**

(ii) $E = mgh$ **or** $E = Wh$ **C1**

$= 2.5 \times 0.32$

$= 0.80$ J **A1**

(b)(i) kinetic energy $= 0.80 - 0.64$ **A1**

$= 0.16$ J

(ii) $E = ½mv^2$ **C1**

$0.16 = ½ \times (2.5 / 9.81) \times v^2$

$v = 1.1$ m s^{-1} **A1**

---

29 (a) the point where (all) the weight (of the body) is taken to act **B1**

(b)(i) vertical component $= 54 \sin 35°$ **A1**

$= 31$ N

(ii) the (line of action of the) force (at B) passes through (point) A **B1**

or

the (line of action of the) force (at B) has zero (perpendicular) distance from (point) A

(iii) $54 \sin 35° \times 0.68$ **or** $54 \cos 35° \times 0.68$ **or** $W \times 0.34$ **C1**

$54 \sin 35° \times 0.68 + 54 \cos 35° \times 0.68 = W \times 0.34$ so $W = 150$ (N) **A1**

(iv) total vertical force $= 150 - 31$ **A1**

$= 120$ N

(c) $(\Delta)E = mg(\Delta)h$ **C1**

$E = ½mv^2$ **C1**

ratio $= (m \times 9.81 \times 4.8) / (½ \times m \times 9.2^2)$ **or** $(9.81 \times 4.8) / (½ \times 9.2^2)$ **C1**

$= 1.1$ **A1**

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30 (a) (i)	**1.**	$W = mas$	**B1**
	2.	$s = (v^2 - u^2) / 2a$	**B1**
(ii)		W/work equals energy transferred/gain or change in kinetic energy	**B1**
		$W (= mas) = ma(v^2 - u^2) / 2a$	**B1**
		leading to $W = m(v^2 - u^2) / 2$ (so KE = $\frac{1}{2}mv^2$)	
(b) (i)	**1.**	solid curved line drawn from X to Y along path of ball and labelled D	**B1**
	2.	solid straight line drawn from X to Y and labelled S	**B1**
(ii)		$(\triangle)E = mg(\triangle)h$	**C1**
		$4.5 = (0.040 \times 9.81 \times h) + (\frac{1}{2} \times 0.040 \times 9.5^2)$	**C1**
		$h = 6.9$ m	**A1**
(iii)		line with a negative gradient starting from a non-zero value of kinetic energy when the vertical height is zero	**M1**
		straight line ends at a non-zero value of kinetic energy when the vertical height is h	**A1**

---

31 (a) $k = F / x$ **or** k = gradient

e.g. $k = 4.0 / 0.050$

$k = 80$ N m^{-1}

(b) $E = \frac{1}{2}Fx$ **or** $E = \frac{1}{2}kx^2$ **or** E = area under graph

$(\triangle)E = (\frac{1}{2} \times 3.2 \times 0.040) - (\frac{1}{2} \times 1.2 \times 0.015) = 0.055$ J

or

$(\triangle)E = (\frac{1}{2} \times 80 \times 0.040^2) - (\frac{1}{2} \times 80 \times 0.015^2) = 0.055$ J

or

$(\triangle)E = \frac{1}{2} \times (1.2 + 3.2) \times 0.025 = 0.055$ J

(c) $(\triangle)E = mg(\triangle)h$

$= 0.122 \times 9.81 \times (0.120 - 0.095)$

$= 0.030$ J

or

$(\triangle)E = W \times (\triangle)h$

$= 1.2 \times 0.025$

$= 0.030$ J

(d) (i) $E = 0.055 - 0.030$ **A1**

$= 0.025$ J

(ii) $E = \frac{1}{2}mv^2$ **C1**

$v = [(2 \times 0.025) / 0.122]^{0.5}$ **A1**

$= 0.64$ m s^{-1}

---

32 (a)	force on body A (by body B) is equal (in magnitude) to force on body B (by body A)	**B1**
	force on body A (by body B) is opposite (in direction) to force on body B (by body A)	**B1**
(b) (i)	$m_X \times 5v$ **or** $(m_X + m_Y) \times v$	**C1**
	$m_X \times 5v = (m_X + m_Y) \times v$ (so) $m_Y / m_X = 4$	**A1**
(ii)	$(E =)\ ½mv^2$	**C1**
	ratio $= [½ \times (m_X + m_Y) \times v^2] / [½ \times m_X \times (5v)^2]$	**C1**
	$= 0.2$	**A1**
(iii)	ratio = 1	**A1**
(c) (i)	**1.** (magnitude of resultant force is) zero	**B1**
	2. (magnitude of resultant force is) constant	**B1**
	(direction of resultant force is) opposite to the momentum	**B1**
(ii)	horizontal line from (0 ms, 0 squares) ending at (20 ms, 0 squares)	**B1**
	straight line from (20 ms, 0 squares) ending at (40 ms, 4.0 squares [= 4.0 cm vertically])	**B1**
	horizontal line from (40 ms, 4.0 squares) ending at (60 ms, 4.0 squares)	**B1**

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33 (a)	resultant force (in any direction) is zero	**B1**
	resultant torque/moment (about any point) is zero	**B1**
(b) (i)	**1.** $T \sin 53° = 2.4$	**A1**
	$T = 3.0$ N	
	2. $F = T \cos 53°$ **or** $F^2 = T^2 - 2.4^2$	**A1**
	$F = 1.8$ N	
(ii)	$\sigma = T / A$ **or** $\sigma = F / A$	**C1**
	$A = \pi d^2 / 4$ **or** $A = \pi r^2$	**C1**
	$\sigma = 3.0 \times 4 / [\pi \times (0.50 \times 10^{-3})^2]$	**A1**
	$= 1.5 \times 10^7$ Pa	
(c) (i)	$h = 75 - 75 \sin 53° = 15$ cm	**A1**
(ii)	$(\Delta)E = mg(\Delta)h$ **or** $(\Delta)E = W(\Delta)h$	**C1**
	$(\Delta)E = 2.4 \times 15 \times 10^{-2}$	**A1**
	$= 0.36$ J	
(iii)	$E = ½mv^2$	**B1**
	$0.36 = ½ \times (2.4 / 9.81) \times v^2$	**C1**
	$v = 1.7$ m s^{-1}	**A1**

34	**(a)**	(work done =) force $\times$ displacement in direction of the force	**B1**
	(b) (i)	**1.** $(\Delta)E = mg(\Delta)h$	**C1**
		$= 0.42 \times 9.81 \times 78$	**A1**
		$= 320$ J	
		2. $E = ½mv^2$	**C1**
		$(\Delta)E = ½ \times 0.42 \times 23^2$	**A1**
		$= 110$ J	
	(ii)	work done = 320 – 110 (= 210 N)	**C1**
		average resistive force = 210 / 78	**A1**
		= 2.7 N	
	(c)	downward sloping line from (0, g) to a non-zero value on the time axis	**M1**
		line is curved with a gradient that becomes less negative **and** the line meets t-axis at time $t < T$	**A1**

35.

2(a)	force × displacement in the direction of the force
(b)(i)	$E = \frac{1}{2}mv^2$
	(m =) $23 \times 2 / 16^2 = 0.18$ (kg)
(b)(ii)	$(\Delta)E = mg(\Delta)h$ $60 = 0.18 \times 9.81 \times h$
	$h = 34$ m
(b)(iii)	(work done =) 60 – 23 = 37 (J)
	average resistive force = 37 / 34 = 1.1 N
2(c)	air resistance (acting on ball) increases
	resultant force (on ball) decreases or weight constant and air resistance increases
	acceleration decreases

36.

(a)(i)	$P = Fv$
	$= 18 \times 1.4$ $= 25$ W
(a)(ii)	$a = F / m$
	$a = 18 / 72 = 0.25$ (m s^{-2}) $t = 1.4 / 0.25$
	= 5.6 s
(b)(i)	$a = (54 – 18) / 72$ **or** $36 / 72$ (= 0.50 m s^{-2})
	$v^2 = 2 \times 0.50 \times 9.5$
	$v = 3.1$ m s^{-1}
(b)(ii)	$W = 54 \times 9.5$ $= 510$ J
b)(iii)	curved line from the origin
	gradient of line increases
2(c)	(force due to) air resistance increases/changes/not constant **or** air resistance increases with speed

37.

3(a)	$(\Delta)E = mg(\Delta)h$ **or** $W(\Delta)h$
	$= 330 \times (4.0 - 1.1)$ $= 960$ J
3(b)	(work =) 960 – 540 (= 420 J)
	distance moved = (960 – 540) / 52 = 8.1 m
(c)(i)	$E = ½mv^2$
	$540 = ½ \times (330 / 9.81) \times v^2$ $v = 5.7$ m s^{-1}
(c)(ii)	speed = horizontal component of velocity $= 5.7 \times \cos 41°$
	$= 4.3$ m s^{-1}

38.

3(a)	work (done) / time (taken)
(b)(i)	zero / 0 J
(b)(ii)	work done $= 440 \times 25$ $= 1.1 \times 10^4$ J
(c)(i)	$(\Delta)E_{(P)} = mg(\Delta)h$
	$h = 4.8 \times 10^4 / (1700 \times 9.81)$ $= 2.9$ m
(c)(ii)	$\theta = \sin^{-1}(2.9 / 25)$ $= 6.7°$
3(d)	work done $= 4.8 \times 10^4 + 1.1 \times 10^4$ (= 5.9×10^4 J)
	time $= 5.9 \times 10^4 / 1.7 \times 10^4$ $= 3.5$ s

39.

4(a)	ratio = 300 / 3200 = 0.094
4(b)	$E = ½mv^2$ or $E \propto v^2$
	ratio = $(0.094)^{0.5}$ = 0.31
4(c)	work (done against frictional force) = 3200 – 300 (=2900)
	length = 2900 / 76 = 38 m
(d)(i)	$E = ½kx^2$ or $E = ½Fx$ <u>and</u> $F = kx$ $140 = ½ \times 63 \times x^2$ or $140 = ½Fx$ <u>and</u> $F = 63x$
	x = 2.1 m
(d)(ii)	percentage efficiency = (140 / 300) × 100 = 47%
d)(iii)	curved line from the origin
	gradient of line increases

40.

(a)(i)	work (done) / time (taken)
(a)(ii)	$W = Fx$ **or** $E = Fx$
	$(P =)\ Fx / t = Fv$ **or** $P = Fvt / t = Fv$
(b)(i)	T = 430 sin 11° **or** 430 cos 79°
	= 82 N
(b)(ii)	speed = 56 / 82
	= 0.68 m s^{-1}
(b)(iii)	no change in kinetic energy (of block)
(b)(iv)	(input power) = 56 / 0.80 (= 70 W)
	time taken = 1.2×10^3 / 70 = 17 s
	or
	(useful energy) = 1200 × 80 / 100 (= 960 J)
	time taken = 960 / 56 = 17 s

41.

4(a)	work (done) = force × displacement
	(force = mg and distance = Δh) (so) work (done) = $mg\Delta h$ <u>and</u> work = $\Delta E_{(P)}$ (so $\Delta E_{(P)} = mg\Delta h$)
4(b)	gravitational potential (energy) to heat/thermal (energy)
(c)(i)	$P = mg(\Delta)h / (\Delta)t$ **or** Fv
	$P = (0.60 \times 9.81 \times 1.4) / 4.0$ **or** $0.60 \times 9.81 \times (1.4 / 4.0)$ $= 2.1$ W
(c)(ii)	$P = I^2R$ **or** IV **or** V^2 / R
	$= 0.09^2 \times 47$ **or** 0.09×4.23 **or** $4.23^2 / 47$ $= 0.38$ W
c)(iii)	efficiency $= P_{out} / P_{in}$ ($\times$ 100) **or** E_{out} / E_{in} ($\times$ 100)
	$= 0.38 / 2.1$ ($\times$ 100) **or** $0.38 \times 4.0 / 2.1 \times 4.0$ ($\times$ 100) $= 0.18$ **or** 18%

42.

(a)(i)	work done per unit time
(a)(ii)	$W = Fs$
	$P = Fs / t$ <u>and</u> (so) $P = Fv$
3(b)	(F =) $130 \times 10^3 / 25 = 5200$ (N)
(c)(i)	(component of weight =) $mg \sin\theta$
(c)(ii)	F (along slope due to weight) $= 36\,000 \times 9.81 \times \sin 1.4°$ (= 8600 N)
	(total) $F = 5200 + 36\,000 \times 9.81 \times \sin 1.4°$ (= 13 800 N)
	$P = 13\,800 \times 25$ $= 350 \times 10^3$ (W) $= 350$ kW

43.

2(a)(i)	$E = \sigma / \varepsilon$ **or** $E = F / A\varepsilon$
	$A = 1.4 \times 10^4 / (2.2 \times 10^{11} \times 0.0012)$ $= 5.3 \times 10^{-5}\ m^2$
(a)(ii)	$(\Delta)h = 0.64 \times 0.49$ (= 0.3136)
	$(\Delta)E = mg(\Delta)h$ **or** $W(\Delta)h$
	$= 1.4 \times 10^4 \times 0.64 \times 0.49$ $= 4.4 \times 10^3$ J
2(b)	$P = Fv$ **or** W / t
	$= (1.4 \times 10^4 \times 0.64) / 0.56$ **or** $(4.4 \times 10^3 / 0.49) / 0.56$
	$= 1.6 \times 10^4$ W
2(c)	$m = 1.4 \times 10^4 / 9.81$ (= 1427 kg)
	(resultant) $F = (1.4 \times 10^4 / 9.81) \times 1.3$ (= 1855 N)
	$T = 1.4 \times 10^4 - 1855$ **or** $(1.4 \times 10^4 / 9.81) \times (9.81 - 1.3)$ $= 1.2 \times 10^4$ N
2(d)	upward sloping straight line from $(t_X, 0)$ to t_Y
	from t_Y to t_Z: an upward sloping curve with decreasing magnitude of gradient (that is horizontal at t_Z)

44.

2(a)	the point where (all) the weight (of the object) is taken to act
(b)(i)	(54×0.45) **or** (2.4×0.95) **or** $(T \sin 30° \times 1.3)$
	$(54 \times 0.45) = (2.4 \times 0.95) + (T \sin 30° \times 1.3)$
	$T = 34$ N
(b)(ii)	resultant moment $= (54 \times 0.45) - (2.4 \times 0.95)$ **or** $(34 \sin 30° \times 1.3)$ $= 22$ N m
(c)(i)	$(\Delta)E = mg(\Delta)h$ **or** $W(\Delta)h$
	$= 2.4 \times 1.8$ $= 4.3$ J

2(c)(ii)	$E = \frac{1}{2}mv^2$
	$= \frac{1}{2} \times (2.4 / 9.81) \times 3.4^2$ $= 1.4$ J (at X)
	kinetic energy at Y $= 4.3 + 1.4$ $= 5.7$ J
	or
	$\frac{1}{2}mv^2 = \frac{1}{2}mu^2 + mg(\Delta)h$
	$v^2 = 3.4^2 + 2 \times 9.81 \times 1.8$ $v^2 = 46.9$ so $v = 6.85$ (m s^{-1}) KE $= \frac{1}{2} \times (2.4 / 9.81) \times 6.85^2$
	$= 5.7$ J
(c)(iii)	no variation **or** acceleration is (always) vertically downwards
(c)(iv)	horizontal straight line at a non-zero value of velocity